Metals

Higher · Materials & manufacture · Worksheet

D&M WITH MR KELLY

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ explain ferrous, non-ferrous and alloy
- ☐ describe the properties and uses of iron, cast iron, mild steel and stainless steel
- ☐ describe the properties and uses of high-speed steel, tin and zinc
- ☐ describe the properties and uses of aluminium, copper and brass

I DO — Worked example

Read how a full-mark answer is built

Two outdoor picnic benches: a park bench (galvanised mild steel frame, recycled HDPE slats, stainless steel bolts) and a garden bench (cast aluminium ends, oak slats).

1 Explain why the materials chosen for these benches are suitable. You should make six valid points. (6 marks)

Full-mark answer each ✓ is one mark

The galvanised mild steel frame is strong enough to carry the weight of several adults ✓, and the zinc coating stops it rusting outdoors ✓. The recycled HDPE slats are weatherproof and will not rot like timber ✓, and because the colour runs through the material, scratches do not show ✓. The stainless steel bolts will not corrode and stain the slats ✓. The oak slats on the garden bench are very durable outdoors and have an attractive grain that suits a garden setting ✓.

Why marks are not awarded

"Mild steel is strong. HDPE is waterproof. Stainless steel is strong. Oak is strong."

1–2 marks: properties are stated, not explained; "strong" repeated scores once.

WE DO — Guided practice

Use the starters and word bank to help you

2 Explain why high-speed steel is used for drill bits and why steel buckets are coated in zinc. (2 marks)

Sentence starters

One reason is … because …

Another reason is … so …

Word bank

coating · High-speed steel · protects · cutting edge · temperatures · galvanising · rusting · scratched

Tip: 2 marks = 2 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) Brass is an alloy of copper and...
☐ Tin ☐ Zinc ☐ Iron ☐ Aluminium

b) Which metal is used to coat steel food cans?
☐ Zinc ☐ Tin ☐ Copper ☐ Brass

c) Which metal suits kitchen cutlery?
☐ Mild steel ☐ Cast iron ☐ Stainless steel ☐ Copper

d) Which metal is ferrous?
☐ Mild steel ☐ Brass ☐ Aluminium ☐ Copper

4 Key terms: write the letter of the matching description next to each term.

1. **Ferrous metal**	____	**A** Copper + zinc alloy: polishes well, corrosion resistant. Taps, fittings.
2. **Alloy**	____	**B** A metal mixed with other elements to improve its properties.
3. **Mild steel**	____	**C** Ductile, malleable, easily welded. Car bodies, nuts and bolts.
4. **Stainless steel**	____	**D** Contains iron; magnetic (except stainless); rusts.
5. **Aluminium**	____	**E** Corrosion resistant and hygienic. Cutlery, sinks.
6. **Brass**	____	**F** Lightweight with good strength-to-weight ratio. Cans, ladders, aircraft.

5 Give the properties and one use of Copper. (2 marks)

6 Give the properties and one use of High-speed steel. (2 marks)

Check and reflect

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?